

Element H

Prototype Testing and Data Collection Plan

A final design was not made so the engineers were unable to proceed further prototype making and testing. However, we did test different containers using humidity sensors. In total we tested six different materials which were an insulated metal bottle, cigar box, zip lock bag, lunch bag, tupperware container, and bubble wrap foil. We found that the wooden cigar box and plastic tupperware container worked best; the humidity inside the containers stayed low while the humidity percentage outside of the containers were constantly high.

To test the prototype, the students plan to place humidity sensors outside and inside the Anti-Humidity Box along with painted paper to mimic artwork from the art students. The most optimal results would be paper drying properly, without getting wet from the humidity and low humidity inside the box overall, along with cool temperatures in the box at all times. The minimum requirement would be the painted papers must dry properly.